CSS Complete Reference Guide

Comprehensive CSS Training Manual

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1. Introduction to CSS

What is CSS?

CSS (Cascading Style Sheets) is a stylesheet language used to describe the presentation of HTML documents. It controls layout, colors, fonts, spacing, and overall visual appearance of web pages.

Key Characteristics:

- **Separation of Concerns**: Keeps styling separate from HTML structure
- **Cascading**: Styles can cascade from multiple sources
- **Specificity**: Rules determine which styles apply
- **Inheritance**: Some properties inherit from parent elements

How CSS Works

1. **Browser loads** HTML document
2. **Browser parses** CSS rules
3. **Browser applies** styles based on specificity
4. **Browser renders** the styled page

CSS History

- **1996**: CSS1 released
- **1998**: CSS2 released
- **2011**: CSS3 modular approach begins
- **Present**: CSS continues to evolve with new features

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2. Adding CSS to HTML

External CSS (Recommended)

HTML:

```
<head>
   <link rel="stylesheet" href="styles.css">
</head>
```

styles.css:

```
body {
   font-family: Arial, sans-serif;
   background-color: #f0f0f0;
}
```

Advantages:

- Reusable across multiple pages
- Better organization
- Cached by browser (faster loading)
- Easier maintenance

Internal CSS

```
<head>
   <style>
      body {
         font-family: Arial, sans-serif;
         background-color: #f0f0f0;
      }

      h1 {
         color: navy;
      }
   </style>
</head>
```

Use When:

- Styles are page-specific

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- Single-page websites
- Quick prototypes

Inline CSS

This is styled inline.

Use When:

- Quick testing only
- Email templates (limited CSS support)

Avoid in production - hard to maintain!

Import Method

@import url('styles.css');
@import url('https://fonts.googleapis.com/css?family=Roboto');

Note: Can impact performance; prefer `<link>` tags

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3. Selectors

Basic Selectors

Element Selector

```
p {
    color: blue;
}
```

Class Selector

```
.classname {
    font-size: 16px;
}

<div class="classname">Content</div>
```

ID Selector

```
#unique-id {
    background: yellow;
}

<div id="unique-id">Content</div>
```

Universal Selector

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}
```

Combinator Selectors

Descendant Selector (space)

```
div p {
    /* All p inside div (any level) */
    color: red;
```

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```
}
```

Child Selector (>)

```
div > p {
    /* Direct children only */
    color: blue;
}
```

Adjacent Sibling (+)

```
h1 + p {
    /* p immediately after h1 */
    font-weight: bold;
}
```

General Sibling (~)

```
h1 ~ p {
    /* All p siblings after h1 */
    color: gray;
}
```

Attribute Selectors

```
/* Has attribute */
[disabled] {
    opacity: 0.5;
}

/* Exact value */
[type="text"] {
    border: 1px solid gray;
}

/* Contains value */
[class*="btn"] {
    padding: 10px;
}

/* Starts with */
[href^="https"] {
```

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```
    color: green;
}

/* Ends with */
[src$=".jpg"] {
    border: 2px solid black;
}

/* Space-separated contains */
[class~="active"] {
    font-weight: bold;
}
```

Pseudo-classes

User Action Pseudo-classes

```
a:hover {
    color: red;
}

a:active {
    color: blue;
}

a:visited {
    color: purple;
}

input:focus {
    border-color: blue;
    outline: none;
}
```

Structural Pseudo-classes

```
li:first-child {
    font-weight: bold;
}

li:last-child {
    border-bottom: none;
}
```

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```
li:nth-child(2) {
    /* Second child */
    color: red;
}

li:nth-child(odd) {
    /* 1, 3, 5, etc. */
    background: lightgray;
}

li:nth-child(even) {
    /* 2, 4, 6, etc. */
    background: white;
}

li:nth-child(3n) {
    /* Every 3rd: 3, 6, 9, etc. */
    color: blue;
}

p:first-of-type {
    font-size: 20px;
}

p:last-of-type {
    margin-bottom: 0;
}

div:not(.exclude) {
    /* All divs except .exclude */
    border: 1px solid black;
}
```

Form Pseudo-classes

```
input:required {
    border-color: red;
}

input:optional {
    border-color: gray;
}
```

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```
input:valid {
    border-color: green;
}

input:invalid {
    border-color: red;
}

input:disabled {
    opacity: 0.5;
}

input:enabled {
    cursor: pointer;
}

input:checked {
    /* For checkboxes/radio */
    background: blue;
}
```

Pseudo-elements

```
/* First letter of element */
p::first-letter {
    font-size: 2em;
    font-weight: bold;
}

/* First line of element */
p::first-line {
    font-variant: small-caps;
}

/* Before element content */
.note::before {
    content: "📝 ";
}

/* After element content */
.link::after {
    content: " →";
}
```

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```
/* Selected text */
::selection {
    background: yellow;
    color: black;
}

/* Placeholder text */
input::placeholder {
    color: lightgray;
    font-style: italic;
}
```

Grouping Selectors

```
/* Multiple selectors, same styles */
h1, h2, h3 {
    font-family: Georgia, serif;
    color: navy;
}

/* Multiple classes on one element */
.button.primary {
    background: blue;
    color: white;
}
```

Specificity

Priority (highest to lowest):

1. Inline styles: `style="..."`
2. IDs: `#id`
3. Classes, attributes, pseudo-classes: `.class`, `[attr]`, `:hover`
4. Elements, pseudo-elements: `div`, `::before`

Calculating Specificity:

```
/* (0, 0, 0, 1) - 1 element */
p { color: black; }

/* (0, 0, 1, 0) - 1 class */
.text { color: blue; }
```

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```
/* (0, 1, 0, 0) - 1 ID */
#header { color: red; }

/* (0, 1, 1, 1) - 1 ID, 1 class, 1 element */
#header .nav li { color: green; }
```

!important (use sparingly!)

```
p {
    color: red !important; /* Overrides everything */
}
```

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4. Colors and Backgrounds

Color Values

```
/* Named colors */
color: red;
color: navy;
color: transparent;

/* Hexadecimal */
color: #ff0000;
color: #f00; /* Shorthand */
color: #ff000080; /* With alpha */

/* RGB */
color: rgb(255, 0, 0);

/* RGBA (with transparency) */
color: rgba(255, 0, 0, 0.5); /* 50% transparent */

/* HSL (Hue, Saturation, Lightness) */
color: hsl(0, 100%, 50%);

/* HSLA (with alpha) */
color: hsla(0, 100%, 50%, 0.5);
```

Background Properties

```
.element {
    /* Background color */
    background-color: #f0f0f0;

    /* Background image */
    background-image: url('image.jpg');

    /* Background repeat */
    background-repeat: no-repeat; /* repeat, repeat-x, repeat-y */

    /* Background position */
    background-position: center center;
    background-position: top left;
    background-position: 50% 50%;
    background-position: 10px 20px;
```

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```
    /* Background size */
    background-size: cover; /* Cover entire area */
    background-size: contain; /* Fit within area */
    background-size: 100% auto;
    background-size: 300px 200px;

    /* Background attachment */
    background-attachment: fixed; /* Fixed during scroll */
    background-attachment: scroll; /* Default */

    /* Shorthand */
    background: #f0f0f0 url('image.jpg') no-repeat center/cover;
}
```

Multiple Backgrounds

```
.element {
    background:
        url('image1.png') top left no-repeat,
        url('image2.png') bottom right no-repeat,
        linear-gradient(to bottom, #fff, #000);
}
```

Gradients

Linear Gradients

```
/* Top to bottom */
background: linear-gradient(to bottom, red, blue);

/* Left to right */
background: linear-gradient(to right, red, blue);

/* Diagonal */
background: linear-gradient(to bottom right, red, blue);

/* With angle */
background: linear-gradient(45deg, red, blue);

/* Multiple colors */
background: linear-gradient(red, yellow, green);

/* With stops */
```

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```
background: linear-gradient(red 0%, yellow 50%, green 100%);
```

Radial Gradients

```
/* Circular */
background: radial-gradient(circle, red, blue);

/* Ellipse (default) */
background: radial-gradient(red, blue);

/* At position */
background: radial-gradient(circle at top left, red, blue);

/* With size */
background: radial-gradient(circle 100px, red, blue);
```

Conic Gradients

```
background: conic-gradient(red, yellow, green, blue, red);
background: conic-gradient(from 90deg, red, blue);
```

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5. Text and Fonts

Font Properties

```
.text {
    /* Font family */
    font-family: Arial, Helvetica, sans-serif;
    font-family: 'Times New Roman', Times, serif;
    font-family: 'Courier New', Courier, monospace;

    /* Font size */
    font-size: 16px;
    font-size: 1.5em; /* Relative to parent */
    font-size: 1.5rem; /* Relative to root */
    font-size: 100%; /* Relative to parent */

    /* Font weight */
    font-weight: normal; /* 400 */
    font-weight: bold; /* 700 */
    font-weight: 100; /* Thin */
    font-weight: 900; /* Black */

    /* Font style */
    font-style: normal;
    font-style: italic;
    font-style: oblique;

    /* Font variant */
    font-variant: normal;
    font-variant: small-caps;

    /* Line height */
    line-height: 1.5; /* Unitless - recommended */
    line-height: 24px;
    line-height: 150%;

    /* Shorthand */
    font: italic bold 16px/1.5 Arial, sans-serif;
}
```

Text Properties

```
.text {
    /* Text color */
```

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```
color: #333;

/* Text alignment */
text-align: left;
text-align: center;
text-align: right;
text-align: justify;

/* Text decoration */
text-decoration: none;
text-decoration: underline;
text-decoration: overline;
text-decoration: line-through;
text-decoration: underline wavy red;

/* Text transform */
text-transform: none;
text-transform: uppercase;
text-transform: lowercase;
text-transform: capitalize;

/* Text indent */
text-indent: 50px;

/* Letter spacing */
letter-spacing: 2px;
letter-spacing: 0.1em;

/* Word spacing */
word-spacing: 5px;

/* White space */
white-space: normal;
white-space: nowrap; /* No wrapping */
white-space: pre; /* Preserve spaces */
white-space: pre-wrap; /* Preserve and wrap */

/* Text overflow */
text-overflow: clip;
text-overflow: ellipsis; /* ... */

/* Word break */
word-break: normal;
word-break: break-all;
```

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```
    word-break: keep-all;

    /* Overflow wrap */
    overflow-wrap: normal;
    overflow-wrap: break-word;
}
```

Text Shadow

```
.text {
    /* x-offset y-offset blur color */
    text-shadow: 2px 2px 4px rgba(0, 0, 0, 0.5);

    /* Multiple shadows */
    text-shadow:
        2px 2px 4px rgba(0, 0, 0, 0.5),
        -2px -2px 4px rgba(255, 255, 255, 0.5);
}
```

Web Fonts

```
/* Google Fonts */
@import
url('https://fonts.googleapis.com/css2?family=Roboto:wght@300;400;700&display=swap');

/* Custom font */
@font-face {
    font-family: 'CustomFont';
    src: url('font.woff2') format('woff2'),
         url('font.woff') format('woff');
    font-weight: normal;
    font-style: normal;
}

body {
    font-family: 'Roboto', sans-serif;
}
```

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6. Box Model

Understanding the Box Model

Every element is a rectangular box with:

1. **Content** - The actual content
2. **Padding** - Space inside border
3. **Border** - Line around padding
4. **Margin** - Space outside border

Box Sizing

```
/* Content-box (default) */
.element {
    box-sizing: content-box;
    width: 300px; /* Content width only */
    padding: 20px; /* Adds to total width */
    border: 5px solid black; /* Adds to total width */
    /* Total width: 300 + 40 + 10 = 350px */
}

/* Border-box (recommended) */
.element {
    box-sizing: border-box;
    width: 300px; /* Total width including padding & border */
    padding: 20px;
    border: 5px solid black;
    /* Total width: 300px */
}

/* Apply to all elements */
* {
    box-sizing: border-box;
}
```

Width and Height

```
.element {
    /* Fixed dimensions */
    width: 300px;
    height: 200px;

    /* Percentage */
```

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```
    width: 50%;
    height: 100%;

    /* Viewport units */
    width: 50vw; /* 50% of viewport width */
    height: 100vh; /* 100% of viewport height */

    /* Min/Max constraints */
    min-width: 200px;
    max-width: 1200px;
    min-height: 100px;
    max-height: 500px;

    /* Auto */
    width: auto; /* Default */
    height: auto;
}
```

Padding

```
.element {
    /* All sides */
    padding: 20px;

    /* Vertical | Horizontal */
    padding: 10px 20px;

    /* Top | Horizontal | Bottom */
    padding: 10px 20px 30px;

    /* Top | Right | Bottom | Left */
    padding: 10px 20px 30px 40px;

    /* Individual sides */
    padding-top: 10px;
    padding-right: 20px;
    padding-bottom: 30px;
    padding-left: 40px;
}
```

Border

```
.element {
```

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```
    /* Shorthand: width style color */
    border: 2px solid black;

    /* Individual properties */
    border-width: 2px;
    border-style: solid;
    border-color: black;

    /* Border styles */
    border-style: solid;
    border-style: dashed;
    border-style: dotted;
    border-style: double;
    border-style: groove;
    border-style: ridge;
    border-style: inset;
    border-style: outset;
    border-style: none;

    /* Individual sides */
    border-top: 2px solid red;
    border-right: 2px solid blue;
    border-bottom: 2px solid green;
    border-left: 2px solid yellow;

    /* Border radius (rounded corners) */
    border-radius: 10px;
    border-radius: 50%; /* Circle */
    border-radius: 10px 20px 30px 40px; /* Each corner */
    border-top-left-radius: 10px;
    border-top-right-radius: 10px;
    border-bottom-right-radius: 10px;
    border-bottom-left-radius: 10px;
}
```

Margin

```
.element {
    /* All sides */
    margin: 20px;

    /* Vertical | Horizontal */
    margin: 10px 20px;
```

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```
    /* Top | Horizontal | Bottom */
    margin: 10px 20px 30px;

    /* Top | Right | Bottom | Left */
    margin: 10px 20px 30px 40px;

    /* Individual sides */
    margin-top: 10px;
    margin-right: 20px;
    margin-bottom: 30px;
    margin-left: 40px;

    /* Center horizontally */
    margin: 0 auto;

    /* Negative margins */
    margin-top: -10px; /* Pull up */
}
```

Outline

```
.element {
    /* Like border but doesn't affect layout */
    outline: 2px solid red;
    outline-offset: 5px; /* Space between element and outline */

    /* Remove default focus outline */
    outline: none; /* Use carefully - provide alternative focus style */
}
```

Box Shadow

```
.element {
    /* x-offset y-offset blur spread color */
    box-shadow: 5px 5px 10px 0 rgba(0, 0, 0, 0.3);

    /* Inset shadow */
    box-shadow: inset 0 0 10px rgba(0, 0, 0, 0.5);

    /* Multiple shadows */
    box-shadow:
        0 2px 4px rgba(0, 0, 0, 0.1),
        0 4px 8px rgba(0, 0, 0, 0.1),
```

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```
        0 8px 16px rgba(0, 0, 0, 0.1);
}
```

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7. Display and Position

Display Property

```
/* Block */
.block {
    display: block;
    /* Takes full width, starts new line */
    width: 100%; /* Can set width/height */
}

/* Inline */
.inline {
    display: inline;
    /* Flows with text, no line breaks */
    /* Cannot set width/height */
}

/* Inline-block */
.inline-block {
    display: inline-block;
    /* Flows inline but can set width/height */
    width: 200px;
    height: 100px;
}

/* None */
.hidden {
    display: none;
    /* Removes element from flow */
}

/* Flex */
.flex-container {
    display: flex;
}

/* Grid */
.grid-container {
    display: grid;
}

/* Table display */
.table {
```

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```
    display: table;
}
.row {
    display: table-row;
}
.cell {
    display: table-cell;
}
```

Visibility

```
/* Hide but keep space */
.invisible {
    visibility: hidden;
}

/* Show */
.visible {
    visibility: visible;
}
```

Position Property

```
/* Static (default) */
.static {
    position: static;
    /* Normal document flow */
    /* top, right, bottom, left have no effect */
}

/* Relative */
.relative {
    position: relative;
    top: 10px; /* Move down 10px from normal position */
    left: 20px; /* Move right 20px */
    /* Other elements not affected */
}

/* Absolute */
.absolute {
    position: absolute;
    top: 0;
    right: 0;
```

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```
    /* Positioned relative to nearest positioned ancestor */
    /* Removed from normal flow */
}

/* Fixed */
.fixed {
    position: fixed;
    bottom: 20px;
    right: 20px;
    /* Positioned relative to viewport */
    /* Stays in place when scrolling */
}

/* Sticky */
.sticky {
    position: sticky;
    top: 0;
    /* Relative until threshold, then fixed */
}
```

Z-Index (Stacking Order)

```
.element {
    position: relative; /* Must have position */
    z-index: 10; /* Higher = on top */
}

.behind {
    position: relative;
    z-index: 1; /* Lower = behind */
}

/* Negative z-index */
.background {
    position: absolute;
    z-index: -1; /* Behind normal flow */
}
```

Float and Clear

```
/* Float */
.float-left {
    float: left;
```

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```
    margin-right: 10px;
}

.float-right {
    float: right;
    margin-left: 10px;
}

/* Clear floats */
.clear {
    clear: both; /* Clear left, right, or both */
}

/* Clearfix hack */
.clearfix::after {
    content: "";
    display: table;
    clear: both;
}
```

Overflow

```
.element {
    overflow: visible; /* Default - content not clipped */
    overflow: hidden; /* Clip content */
    overflow: scroll; /* Always show scrollbars */
    overflow: auto; /* Scrollbars only when needed */

    /* Separate axes */
    overflow-x: auto;
    overflow-y: hidden;
}
```

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8. Flexbox

Flex Container Properties

```
.flex-container {
    display: flex; /* or inline-flex */

    /* Direction */
    flex-direction: row; /* Default: left to right */
    flex-direction: row-reverse; /* Right to left */
    flex-direction: column; /* Top to bottom */
    flex-direction: column-reverse; /* Bottom to top */

    /* Wrapping */
    flex-wrap: nowrap; /* Default: single line */
    flex-wrap: wrap; /* Multiple lines */
    flex-wrap: wrap-reverse; /* Wrap in reverse */

    /* Shorthand for direction and wrap */
    flex-flow: row wrap;

    /* Main axis alignment (horizontal if row) */
    justify-content: flex-start; /* Default: start */
    justify-content: flex-end; /* End */
    justify-content: center; /* Center */
    justify-content: space-between; /* Space between items */
    justify-content: space-around; /* Space around items */
    justify-content: space-evenly; /* Equal space */

    /* Cross axis alignment (vertical if row) */
    align-items: stretch; /* Default: stretch to fill */
    align-items: flex-start; /* Top */
    align-items: flex-end; /* Bottom */
    align-items: center; /* Center */
    align-items: baseline; /* Align baselines */

    /* Multiple line alignment */
    align-content: stretch; /* Default */
    align-content: flex-start;
    align-content: flex-end;
    align-content: center;
    align-content: space-between;
    align-content: space-around;
```

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```
    /* Gap between items */
    gap: 20px; /* Both row and column */
    row-gap: 10px;
    column-gap: 20px;
}
```

Flex Item Properties

```
.flex-item {
    /* Grow factor */
    flex-grow: 0; /* Default: don't grow */
    flex-grow: 1; /* Grow to fill space */
    flex-grow: 2; /* Grow twice as much */

    /* Shrink factor */
    flex-shrink: 1; /* Default: shrink if needed */
    flex-shrink: 0; /* Don't shrink */

    /* Base size */
    flex-basis: auto; /* Default: based on content */
    flex-basis: 200px; /* Fixed size */
    flex-basis: 30%; /* Percentage */

    /* Shorthand: grow shrink basis */
    flex: 1; /* flex: 1 1 0 */
    flex: 0 0 200px; /* Fixed 200px */
    flex: 1 1 auto; /* Default */

    /* Individual alignment */
    align-self: auto; /* Default: inherit from container */
    align-self: flex-start;
    align-self: flex-end;
    align-self: center;
    align-self: stretch;

    /* Order */
    order: 0; /* Default: source order */
    order: 1; /* Move to end */
    order: -1; /* Move to start */
}
```

Flexbox Examples

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Horizontal Navigation

```
.nav {
    display: flex;
    justify-content: space-between;
    align-items: center;
}
```

Card Layout

```
.card-container {
    display: flex;
    flex-wrap: wrap;
    gap: 20px;
}

.card {
    flex: 1 1 300px; /* Grow, shrink, min 300px */
}
```

Centered Content

```
.center {
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
}
```

Holy Grail Layout

```
.container {
    display: flex;
    flex-direction: column;
    min-height: 100vh;
}

.header,
.footer {
    flex-shrink: 0;
}
```

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```
.main {
    display: flex;
    flex: 1;
}

.sidebar {
    flex: 0 0 200px;
}

.content {
    flex: 1;
}
```

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9. CSS Grid

Grid Container Properties

```
.grid-container {
    display: grid; /* or inline-grid */

    /* Define columns */
    grid-template-columns: 200px 200px 200px; /* Fixed */
    grid-template-columns: 1fr 1fr 1fr; /* Fractions */
    grid-template-columns: repeat(3, 1fr); /* Repeat */
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr)); /* Responsive */
    grid-template-columns: 200px auto 200px; /* Mixed */

    /* Define rows */
    grid-template-rows: 100px 200px 100px;
    grid-template-rows: repeat(3, 100px);
    grid-template-rows: auto 1fr auto;

    /* Gaps */
    gap: 20px; /* Both row and column */
    row-gap: 10px;
    column-gap: 20px;

    /* Alignment */
    justify-items: stretch; /* Default: fill cell */
    justify-items: start; /* Left */
    justify-items: end; /* Right */
    justify-items: center; /* Center */

    align-items: stretch; /* Default */
    align-items: start; /* Top */
    align-items: end; /* Bottom */
    align-items: center; /* Center */

    /* Grid alignment within container */
    justify-content: start; /* Default */
    justify-content: end;
    justify-content: center;
    justify-content: space-between;
    justify-content: space-around;
    justify-content: space-evenly;

    align-content: start;
```

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```
    align-content: end;
    align-content: center;
    align-content: space-between;
    align-content: space-around;
    align-content: space-evenly;
}
```

Grid Template Areas

```
.grid-container {
    display: grid;
    grid-template-areas:
        "header header header"
        "sidebar main main"
        "sidebar main main"
        "footer footer footer";
    grid-template-columns: 200px 1fr 1fr;
    grid-template-rows: auto 1fr 1fr auto;
    gap: 10px;
}

.header {
    grid-area: header;
}

.sidebar {
    grid-area: sidebar;
}

.main {
    grid-area: main;
}

.footer {
    grid-area: footer;
}
```

Grid Item Properties

```
.grid-item {
    /* Column span */
    grid-column: 1 / 3; /* Start / End */
    grid-column: 1 / span 2; /* Start / Span */
```

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```
    grid-column: span 2; /* Span from auto */

    /* Row span */
    grid-row: 1 / 3;
    grid-row: 2 / span 2;

    /* Shorthand */
    grid-area: 1 / 1 / 3 / 3; /* row-start / col-start / row-end / col-end */

    /* Individual alignment */
    justify-self: start; /* Left */
    justify-self: end; /* Right */
    justify-self: center; /* Center */
    justify-self: stretch; /* Fill */

    align-self: start; /* Top */
    align-self: end; /* Bottom */
    align-self: center; /* Center */
    align-self: stretch; /* Fill */
}
```

Implicit Grid

```
.grid-container {
    display: grid;
    grid-template-columns: repeat(3, 1fr);

    /* Control auto-generated rows */
    grid-auto-rows: 100px;
    grid-auto-rows: minmax(100px, auto);

    /* Control auto-generated columns */
    grid-auto-columns: 200px;

    /* Flow direction */
    grid-auto-flow: row; /* Default: fill rows */
    grid-auto-flow: column; /* Fill columns */
    grid-auto-flow: dense; /* Fill gaps */
}
```

Grid Examples

Responsive Image Gallery

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```
.gallery {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 20px;
}
```

Dashboard Layout

```
.dashboard {
    display: grid;
    grid-template-columns: 250px 1fr;
    grid-template-rows: 60px 1fr 40px;
    grid-template-areas:
        "sidebar header"
        "sidebar main"
        "sidebar footer";
    gap: 10px;
    min-height: 100vh;
}
```

Card Grid with Different Sizes

```
.card-grid {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 20px;
}

.card-large {
    grid-column: span 2;
    grid-row: span 2;
}

.card-wide {
    grid-column: span 2;
}

.card-tall {
    grid-row: span 2;
}
```

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10. Responsive Design

Mobile-First Approach

```
/* Base styles (mobile) */
.container {
    width: 100%;
    padding: 15px;
}

/* Tablet (768px and up) */
@media (min-width: 768px) {
    .container {
        width: 750px;
        margin: 0 auto;
    }
}

/* Desktop (1024px and up) */
@media (min-width: 1024px) {
    .container {
        width: 960px;
    }
}

/* Large Desktop (1440px and up) */
@media (min-width: 1440px) {
    .container {
        width: 1200px;
    }
}
```

Media Query Syntax

```
/* Width */
@media (min-width: 768px) { }
@media (max-width: 1023px) { }
@media (min-width: 768px) and (max-width: 1023px) { }

/* Height */
@media (min-height: 600px) { }

/* Orientation */
@media (orientation: portrait) { }
```

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```
@media (orientation: landscape) { }

/* Resolution */
@media (min-resolution: 2dppx) { } /* Retina */

/* Hover capability */
@media (hover: hover) { }

/* Print */
@media print { }

/* Dark mode */
@media (prefers-color-scheme: dark) { }

/* Reduced motion */
@media (prefers-reduced-motion: reduce) { }

/* Multiple conditions */
@media (min-width: 768px) and (max-width: 1024px) and (orientation: portrait) { }

/* OR condition */
@media (max-width: 767px), (min-width: 1440px) { }
```

Responsive Units

```
.element {
    /* Relative to parent font size */
    font-size: 1.5em;
    padding: 2em;

    /* Relative to root (html) font size */
    font-size: 1.5rem;
    margin: 2rem;

    /* Percentage */
    width: 50%;

    /* Viewport width */
    width: 50vw; /* 50% of viewport width */
    font-size: 5vw;

    /* Viewport height */
    height: 100vh; /* 100% of viewport height */
    min-height: 50vh;
```

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```
    /* Viewport minimum */
    font-size: 3vmin; /* 3% of smaller dimension */

    /* Viewport maximum */
    font-size: 5vmax; /* 5% of larger dimension */
}
```

Responsive Typography

```
/* Fluid typography */
body {
    font-size: calc(16px + (24 - 16) * ((100vw - 320px) / (1920 - 320)));
}

/* Using clamp */
.heading {
    font-size: clamp(1.5rem, 5vw, 3rem);
    /* min, preferred, max */
}
```

Responsive Images

```
/* Basic responsive image */
img {
    max-width: 100%;
    height: auto;
}

/* Picture element (in HTML) */
/*
<picture>
    <source media="(min-width: 1024px)" srcset="large.jpg">
    <source media="(min-width: 768px)" srcset="medium.jpg">
    <img src="small.jpg" alt="Description">
</picture>
*/

/* Object fit */
.image {
    width: 100%;
    height: 300px;
    object-fit: cover; /* cover, contain, fill, none, scale-down */
```

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```
    object-position: center;
}
```

Container Queries (New!)

```
/* Define container */
.card-container {
    container-type: inline-size;
    container-name: card;
}

/* Query container */
@container card (min-width: 400px) {
    .card-title {
        font-size: 2rem;
    }
}
```

Responsive Layout Patterns

Column Drop

```
.container {
    display: flex;
    flex-wrap: wrap;
}

.column {
    flex: 1 1 300px;
}
```

Mostly Fluid

```
.container {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 20px;
}
```

Layout Shifter

```
.container {
```

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```
    display: grid;
    grid-template-areas: "main";
}

@media (min-width: 768px) {
    .container {
        grid-template-areas: "sidebar main";
        grid-template-columns: 200px 1fr;
    }
}
```

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11. Transitions and Animations

CSS Transitions

```
.element {
    /* Transition property */
    transition-property: all; /* or specific: background, transform */

    /* Transition duration */
    transition-duration: 0.3s;

    /* Timing function */
    transition-timing-function: ease; /* ease, linear, ease-in, ease-out, ease-in-out */
    transition-timing-function: cubic-bezier(0.1, 0.7, 1.0, 0.1); /* Custom */

    /* Delay */
    transition-delay: 0.1s;

    /* Shorthand: property duration timing-function delay */
    transition: all 0.3s ease 0s;

    /* Multiple properties */
    transition:
        background 0.3s ease,
        transform 0.5s ease-in-out,
        opacity 0.2s linear;
}

/* Example: Hover effect */
.button {
    background: blue;
    color: white;
    transform: scale(1);
    transition: all 0.3s ease;
}

.button:hover {
    background: darkblue;
    transform: scale(1.05);
}
```

Timing Functions

```
.element {
```

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```
    /* Keywords */
    transition-timing-function: linear; /* Constant speed */
    transition-timing-function: ease; /* Default: slow-fast-slow */
    transition-timing-function: ease-in; /* Slow start */
    transition-timing-function: ease-out; /* Slow end */
    transition-timing-function: ease-in-out; /* Slow start and end */

    /* Cubic bezier */
    transition-timing-function: cubic-bezier(0.42, 0, 0.58, 1);

    /* Steps */
    transition-timing-function: steps(4, end);
}
```

CSS Animations

```
/* Define keyframes */
@keyframes slideIn {
    from {
        transform: translateX(-100%);
        opacity: 0;
    }
    to {
        transform: translateX(0);
        opacity: 1;
    }
}

/* With percentages */
@keyframes bounce {
    0% {
        transform: translateY(0);
    }
    50% {
        transform: translateY(-20px);
    }
    100% {
        transform: translateY(0);
    }
}

/* Apply animation */
.element {
    /* Animation name */
```

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```
    animation-name: slideIn;

    /* Duration */
    animation-duration: 1s;

    /* Timing function */
    animation-timing-function: ease-in-out;

    /* Delay */
    animation-delay: 0.5s;

    /* Iteration count */
    animation-iteration-count: 1; /* or infinite */

    /* Direction */
    animation-direction: normal; /* normal, reverse, alternate, alternate-reverse */

    /* Fill mode */
    animation-fill-mode: forwards; /* none, forwards, backwards, both */

    /* Play state */
    animation-play-state: running; /* running or paused */

    /* Shorthand */
    animation: slideIn 1s ease-in-out 0.5s forwards;
}
```

Transform

```
.element {
    /* 2D Transforms */
    transform: translate(50px, 100px); /* Move */
    transform: translateX(50px); /* Move horizontally */
    transform: translateY(100px); /* Move vertically */

    transform: rotate(45deg); /* Rotate */

    transform: scale(1.5); /* Scale uniformly */
    transform: scaleX(2); /* Scale horizontally */
    transform: scaleY(0.5); /* Scale vertically */

    transform: skew(10deg, 5deg); /* Skew */
    transform: skewX(10deg);
    transform: skewY(5deg);
```

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```
    /* 3D Transforms */
    transform: translateZ(50px);
    transform: translate3d(50px, 100px, 75px);

    transform: rotateX(45deg);
    transform: rotateY(45deg);
    transform: rotateZ(45deg);
    transform: rotate3d(1, 1, 1, 45deg);

    transform: perspective(500px) rotateY(45deg);

    /* Multiple transforms */
    transform: translate(50px, 100px) rotate(45deg) scale(1.2);

    /* Transform origin */
    transform-origin: center center; /* Default */
    transform-origin: top left;
    transform-origin: 50% 50%;
    transform-origin: 100px 200px;
}
```

Animation Examples

Fade In

```
@keyframes fadeIn {
    from { opacity: 0; }
    to { opacity: 1; }
}

.fade-in {
    animation: fadeIn 0.5s ease-in;
}
```

Slide In

```
@keyframes slideInLeft {
    from {
        transform: translateX(-100%);
    }
    to {
        transform: translateX(0);
```

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```
    }
}
```

Pulse

```
@keyframes pulse {
    0%, 100% {
        transform: scale(1);
    }
    50% {
        transform: scale(1.1);
    }
}
```

Spin

```
@keyframes spin {
    from {
        transform: rotate(0deg);
    }
    to {
        transform: rotate(360deg);
    }
}

.spinner {
    animation: spin 1s linear infinite;
}
```

Loading Dots

```
@keyframes loading {
    0%, 100% {
        opacity: 0;
    }
    50% {
        opacity: 1;
    }
}

.dot {
    animation: loading 1.4s infinite;
```

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```
}

.dot:nth-child(2) {
    animation-delay: 0.2s;
}

.dot:nth-child(3) {
    animation-delay: 0.4s;
}
```

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12. CSS Variables (Custom Properties)

Defining Variables

```
:root {
    /* Global variables */
    --primary-color: #3498db;
    --secondary-color: #2ecc71;
    --text-color: #333;
    --bg-color: #fff;
    --font-size: 16px;
    --spacing: 20px;
    --border-radius: 8px;
    --transition: all 0.3s ease;
}

/* Scoped variables */
.card {
    --card-padding: 20px;
    --card-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
}
```

Using Variables

```
.element {
    color: var(--primary-color);
    font-size: var(--font-size);
    padding: var(--spacing);
    border-radius: var(--border-radius);
    transition: var(--transition);

    /* Fallback value */
    color: var(--undefined-color, #000);

    /* Calc with variables */
    margin: calc(var(--spacing) * 2);
}
```

Variable Inheritance

```
:root {
    --main-color: blue;
}
```

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```
.parent {
    --main-color: red; /* Override for this scope */
}

.child {
    color: var(--main-color); /* Inherits from parent */
}
```

Variables with Media Queries

```
:root {
    --container-width: 1200px;
    --font-size: 16px;
}

@media (max-width: 768px) {
    :root {
        --container-width: 100%;
        --font-size: 14px;
    }
}

.container {
    max-width: var(--container-width);
    font-size: var(--font-size);
}
```

Color Theming Example

```
:root {
    --primary: #3498db;
    --background: #ffffff;
    --text: #333333;
}

[data-theme="dark"] {
    --primary: #5dade2;
    --background: #1a1a1a;
    --text: #f0f0f0;
}

body {
```

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```
    background-color: var(--background);
    color: var(--text);
}

.button {
    background-color: var(--primary);
}
```

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13. Advanced CSS

Filters

```
.element {
    filter: blur(5px);
    filter: brightness(1.5);
    filter: contrast(200%);
    filter: grayscale(100%);
    filter: hue-rotate(90deg);
    filter: invert(100%);
    filter: opacity(50%);
    filter: saturate(200%);
    filter: sepia(100%);
    filter: drop-shadow(5px 5px 10px rgba(0, 0, 0, 0.5));

    /* Multiple filters */
    filter: blur(2px) brightness(1.2) contrast(110%);
}
```

Backdrop Filter

```
.glass-effect {
    background: rgba(255, 255, 255, 0.1);
    backdrop-filter: blur(10px);
}
```

Blend Modes

```
.element {
    background-blend-mode: multiply;
    /* multiply, screen, overlay, darken, lighten,
       color-dodge, color-burn, hard-light, soft-light,
       difference, exclusion, hue, saturation, color, luminosity */
}

.overlay {
    mix-blend-mode: overlay;
}
```

Clip Path

```
.element {
```

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```
    /* Circle */
    clip-path: circle(50%);

    /* Ellipse */
    clip-path: ellipse(50% 25%);

    /* Polygon */
    clip-path: polygon(50% 0%, 100% 50%, 50% 100%, 0% 50%);

    /* Inset */
    clip-path: inset(10px 20px 30px 40px);
}
```

Multi-column Layout

```
.columns {
    column-count: 3;
    column-gap: 40px;
    column-rule: 1px solid #ddd;
    column-width: 200px;
}

.no-break {
    break-inside: avoid; /* Don't break element */
    page-break-inside: avoid; /* Older browsers */
}
```

Aspect Ratio

```
.element {
    aspect-ratio: 16 / 9;
    aspect-ratio: 1; /* Square */
}
```

Scroll Behavior

```
html {
    scroll-behavior: smooth;
}

.scroll-snap-container {
    scroll-snap-type: y mandatory;
    overflow-y: scroll;
```

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```
}

.scroll-snap-item {
    scroll-snap-align: start;
}
```

Counters

```
body {
    counter-reset: section;
}

h2::before {
    counter-increment: section;
    content: "Section " counter(section) ": ";
}
```

Attribute Content

```
a[href^="https"]::after {
    content: " 🔗";
}

abbr[title]::after {
    content: " (" attr(title) ")";
}
```

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14. CSS Best Practices

Organization

✅ DO:

- Use meaningful class names
- Follow naming conventions (BEM, SMACSS)
- Group related properties
- Comment complex sections
- Use consistent formatting
- Order properties logically

```
/* Good */
.card {
    /* Layout */
    display: flex;
    flex-direction: column;

    /* Box model */
    width: 300px;
    padding: 20px;
    margin: 10px;

    /* Visual */
    background: white;
    border: 1px solid #ddd;
    border-radius: 8px;
    box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
}
```

Performance

✅ DO:

- Minimize use of expensive properties
- Use transform for animations
- Avoid layout thrashing
- Use will-change carefully
- Optimize selectors

```
/* Good - use transform */
.element {
```

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```
    transition: transform 0.3s;
}
.element:hover {
    transform: translateY(-5px);
}

/* Avoid - causes layout */
.element {
    transition: top 0.3s;
}
.element:hover {
    top: -5px;
}
```

Maintainability

✅ **DO:**

- Use CSS variables for reusable values
- Keep specificity low
- Avoid !important
- Write modular, reusable code
- Use consistent units

Responsive Design

✅ **DO:**

- Mobile-first approach
- Use relative units
- Test on multiple devices
- Use flexbox and grid
- Optimize images

Accessibility

✅ **DO:**

- Ensure sufficient color contrast
- Don't rely only on color
- Maintain focus styles
- Support reduced motion
- Use semantic HTML

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```
/* Respect user preferences */
@media (prefers-reduced-motion: reduce) {
    * {
        animation-duration: 0.01ms !important;
        transition-duration: 0.01ms !important;
    }
}
```

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15. CSS Cheat Sheet

Selectors

element { } Element
.class { } Class
#id { } ID
* { } Universal
element, element { } Multiple
element element { } Descendant
element > element { } Child
element + element { } Adjacent sibling
element ~ element { } General sibling

Box Model

width, height
padding
margin
border
box-sizing: border-box;

Display & Position

display: block | inline | inline-block | flex | grid | none
position: static | relative | absolute | fixed | sticky
top, right, bottom, left
z-index

Flexbox

display: flex;
flex-direction: row | column
justify-content: flex-start | center | space-between
align-items: stretch | center | flex-start | flex-end
gap

Grid

display: grid;
grid-template-columns: 1fr 1fr 1fr
grid-template-rows: auto
gap

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grid-column: 1 / 3
grid-row: 1 / 2

Typography

font-family
font-size
font-weight
font-style
line-height
text-align
text-decoration
text-transform
letter-spacing

Colors

color: #hex | rgb() | rgba() | hsl() | hsla()
background-color
background-image
background: gradient

Common Properties

opacity: 0-1
overflow: visible | hidden | scroll | auto
cursor: pointer | default | none
transition: property duration timing-function
animation: name duration timing-function
transform: translate() | rotate() | scale()

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Resources

Documentation:

- MDN CSS: https://developer.mozilla.org/en-US/docs/Web/CSS
- CSS Tricks: https://css-tricks.com
- Can I Use: https://caniuse.com

Tools:

- CSS Validator: https://jigsaw.w3.org/css-validator/
- Flexbox Froggy: https://flexboxfroggy.com
- Grid Garden: https://cssgridgarden.com

Generators:

- Gradient Generator
- Box Shadow Generator
- CSS Grid Generator

END OF CSS GUIDE

Master these concepts through practice and experimentation!

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